

```
import pandas as pd
from sklearn.feature_extraction.text import TfidfVectorizer
from sklearn.metrics.pairwise import cosine_similarity
```

```
df=pd.read_csv("/content/movies.csv")
df.head()
```

	homepage	id	keywords	original_language	original_title	overview
	http://www.avatarmovie.com/	19995	culture clash future space war space colony so...	en	Avatar	In t 22 century paraple Marine d
	go.com/disney pictures/pirates/	285	ocean drug abuse exotic island east india trad...	en	Pirates of the Caribbean: At World's End	Capt Barbos lo believ to dea ha
	ypictures.com/movies/spectre/	206647	spy based on novel secret agent sequel mi6	en	Spectre	A cryp messa fro Boni pa sends h c
	://www.thedarkknightrises.com/	49026	dc comics crime fighter terrorist secret ident...	en	The Dark Knight Rises	Followi the dea of Distr Attorn Harve
	/movies.disney.com/john-carter	49529	based on novel mars medallion space travel pri...	en	John Carter	Jo Carter is w: wea forr milita ce

```
#preprocessing step
df["genres"]=df["genres"].fillna("")
df["overview"]=df["overview"].fillna("")
df["content"]=df["genres"]+" "+df['overview']
```

```
#TF-IDF Vectorizer
tfidf=TfidfVectorizer(stop_words="english")
tfidf_matrix=tfidf.fit_transform(df["content"])
```

```
cosine_sim=cosine_similarity(tfidf_matrix,tfidf_matrix)
```

```
#dropping the duplicates
indices=pd.Series(df.index,index=df["original_title"]).drop_duplicates()
```

```
def recommendation_movie(title,topn=10):
    if title not in indices:
        return "Movie is not found!"
    idx=indices[title]
    sim_scores=list(enumerate(cosine_sim[idx]))
    sim_scores=sorted(sim_scores,key=lambda x:x[1],reverse=True)
    sim_scores=sim_scores[1:topn+1]
    movie_indices=[i[0] for i in sim_scores]
```

```
return df["original_title"].iloc[movie_indices]
```

```
recommendation_movie("Inception")
```

	original_title
2897	Cypher
4401	The Helix... Loaded
134	Mission: Impossible - Rogue Nation
1786	Flatliners
2389	Renaissance
1930	Stone Cold
914	Central Intelligence
322	The Fifth Element
1271	Pandorum
1267	Duplex

dtype: object

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